JAVA PROGRAMMING PROJECT PRESENTATION

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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

Project Title: Warehouse Inventory Monitoring and Stock Reorder System

Course: Mini Project

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INTRODUCTION

The Warehouse Inventory Monitoring and Stock Reorder System in Java is a software application designed to efficiently manage, track, and control inventory levels within a warehouse environment. In today's fast-paced business world, maintaining accurate and up-to-date inventory records is essential for ensuring smooth operations and meeting customer demands. This system leverages the capabilities of Java to provide a reliable and scalable solution for handling inventory-related tasks.

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OBJECTIVE

The main objective of the Warehouse Inventory Monitoring and Stock Reorder System in Java is to design and develop an efficient, reliable, and user-friendly application that automates the process of inventory tracking and management within a warehouse. The system aims to continuously monitor stock levels, accurately record the movement of goods, and maintain up-to-date inventory data with minimal human intervention.

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PROBLEM STATEMENT

Manual inventory management in warehouses often leads to errors, stock shortages, and overstocking due to lack of real-time tracking and automation. A system is needed to efficiently monitor inventory and automate stock reordering to ensure accuracy and optimal stock levels.

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SYSTEM REQUIREMENTS

The hardware requirements for the Warehouse Inventory Monitoring and Stock Reorder System in Java include an Intel i3 or higher processor, a minimum of 4 GB RAM, and at least 500 MB of free storage space, along with standard input/output devices.

The software requirements include Windows, Linux, or macOS as the operating system, Java with JDK 8 or above, an IDE such as Eclipse or IntelliJ IDEA, and a database like MySQL or SQLite with JDBC support for connectivity.

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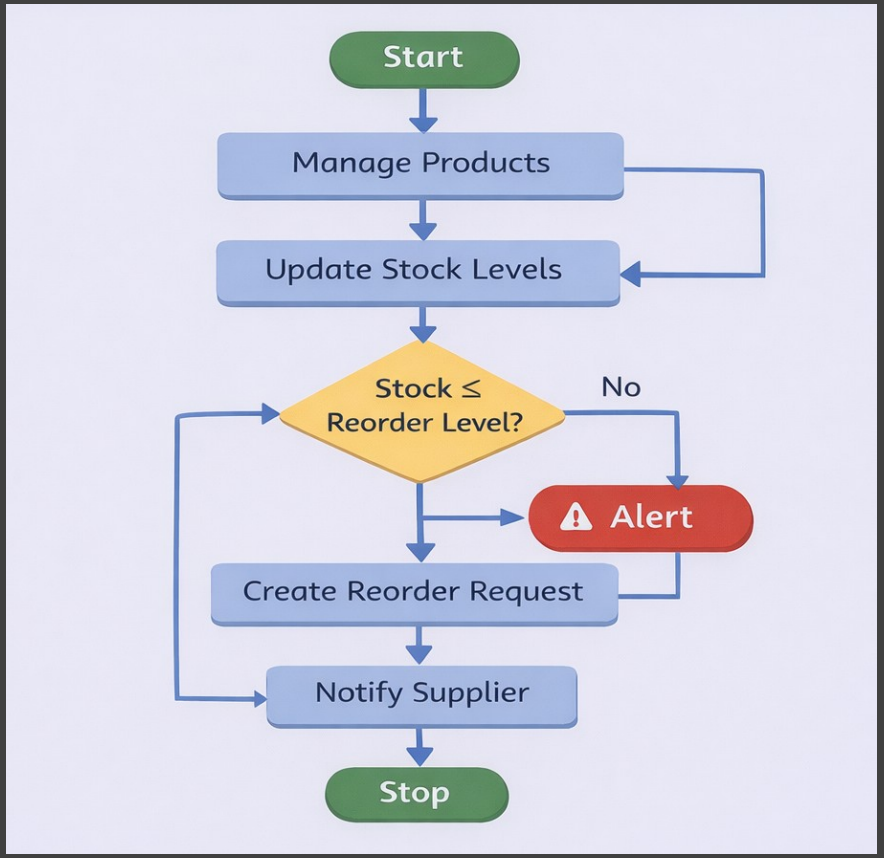
SYSTEM DESIGN



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MODULE DESCRIPTION

The Warehouse Inventory Monitoring and Stock Reorder System in Java consists of four modules: Input, Processing, Output, and File Handling. The Input Module collects product and stock details from the user. The Processing Module updates inventory and checks if stock is below the reorder level. The Output Module displays stock status and alerts. The File Handling Module stores and retrieves data for future use.

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ALGORITHM

1) Start the program
2) Initialize variables (product name, quantity, reorder level)
3) Input product details from the user
4) Update stock quantity
5) Check if stock ≤ reorder level
6) If yes, display "Reorder Required"
7) Else, display "Stock is sufficient"
8) Display current stock details
9) End the program

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IMPLEMENTATION

The Warehouse Inventory Monitoring and Stock Reorder System is implemented using Java classes such as Product and Inventory. It uses ArrayList to store product details and takes user input through the Scanner class. The system uses conditional statements (if-else) to check if the stock is below the reorder level and generates alerts when needed. Loops are used to monitor all products continuously, and methods like addProduct(), updateStock(), and checkReorder() help organize the program. File handling using FileWriter and BufferedReader is used to save and retrieve data, ensuring efficient inventory management.

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OUTPUT SCREENS

```
----- Warehouse Inventory System -----

Enter product name: Rice
Enter quantity: 50
Enter reorder level: 20

Stock is sufficient.
Data saved successfully.
---------------------------------

Enter product name: Sugar
Enter quantity: 10
Enter reorder level: 15

⚠️ Reorder Required!
Data saved successfully.
---------------------------------

Current Inventory:

Product: Rice    Quantity: 50
Product: Sugar   Quantity: 10 (Low Stock)
```

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RESULTS & DISCUSSION

The system successfully accepts product details, updates stock, and generates alerts when stock falls below the reorder level. The output matches the expected results, showing correct inventory monitoring and reorder functionality. The system provides accurate results and performs efficiently for small datasets. Some challenges like managing multiple products, avoiding data loss, and continuous monitoring were faced, which were solved using ArrayList, file handling, and loops. Overall, the system works effectively for basic inventory management.

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CONCLUSION & FUTURE ENHANCEMENT

The Warehouse Inventory Monitoring and Stock Reorder System developed in Java successfully tracks stock levels and automates reordering, reducing errors and improving efficiency. It also enhances understanding of core Java concepts and system design.

Future Enhancements:

The system can be improved by adding a GUI for better usability, integrating a database for efficient data management, and extending features like notifications, reports, and scalability for larger warehouse operations.

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REFERENCES

Java: The Complete Reference- McGraw-Hill ,Herbert Schildt

Oracle Java Documentation for core Java concepts

Tutorials from GeeksforGeeks for inventory system implementation

THANK YOU